

Waves & Oscillation

Waves



The Nature of Waves

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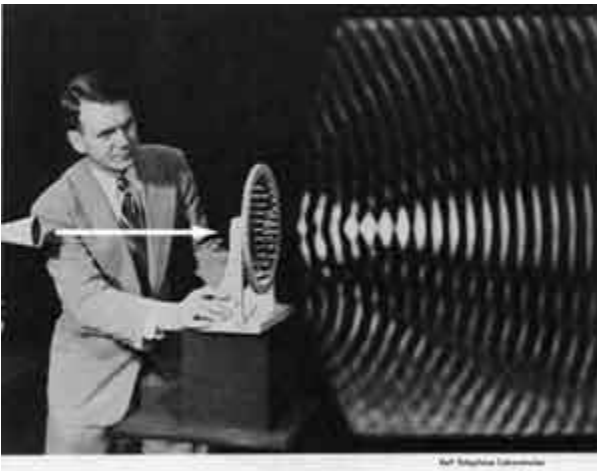
Wave is a traveling disturbance.

Wave carries energy from place to place.

There are two basic types of waves:
Transverse and longitudinal waves.

What is a wave

A **WAVE** is a vibration or disturbance in space.



A **MEDIUM** is the substance that all SOUND WAVES travel through and need to have in order to move.



Different Types of Waves:

Waves can be classified into three types:

- **Mechanical waves**
- **Electromagnetic waves**
- **Matter waves.**

Mechanical Waves act as the propagation of a disturbance through a material medium due to the repeated periodic motion of the particles of the medium about their mean positions, the disturbance being handed over from one particle to the next.

Electromagnetic Waves are the disturbance, which does not require any material medium for its propagation and can travel even through vacuum. They are caused due to varying electric and magnetic fields.

Matter Waves are the waves produced in electrons and particles.

Mechanical Waves:

The existence of medium is essential for propagation. The Method of energy propagation in which disturbance propagates with definite velocity without changing its form is called **Mechanical Wave**.

Energy and momentum propagates by motion of particles of medium. But medium remains at previous position. The mass transfer does not take place here. The Propagation is possible due to property of medium like **elasticity** and **inertia**. Examples: vibration of string, vibration of string, the surface wave produced on the surface of solid and liquid, sound waves, tsunami waves, earthquake P-waves, ultra sounds, vibrations in gas, and oscillations in spring, internal water waves, and waves in slink etc.

Mechanical waves are of two types:

- **Transverse Wave**
- **Longitudinal Wave.**

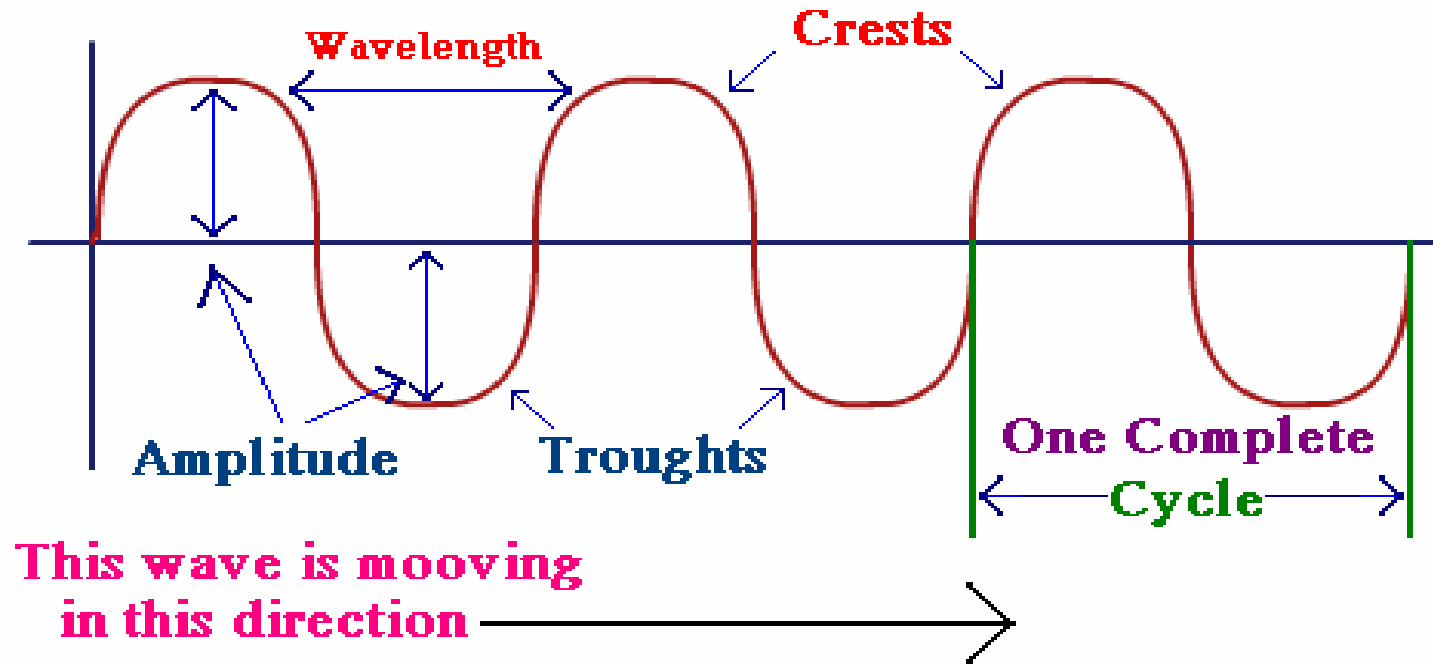
Transverse Wave:

The medium has particles that vibrate in a direction perpendicular to the direction of the propagation of wave. This Kind of wave produced is called **Transverse Wave**.

Transverse waves

wave particles vibrate
in an up-and-down motion.



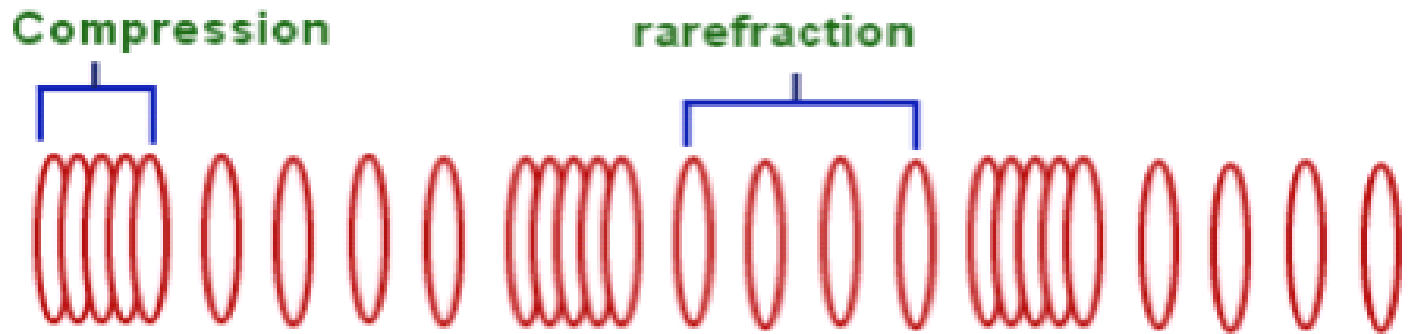


Here the formation of crest and trough takes place. Here the direction of propagation of energy is perpendicular to the direction of oscillations. There are always two directions that are independent of each other that can be used as the direction of wave.

For example: vibration of string.

Longitudinal Waves:

Consider a wave moving. if the vibration of the particles of the medium are in the direction of wave propagation. We call it as **Longitudinal Waves**.



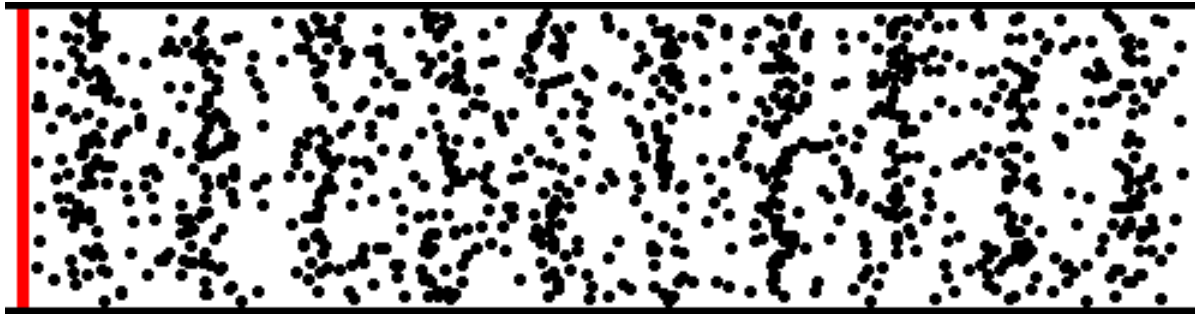
A Longitudinal wave proceeds in the form of compression and rarefaction which is the stretched rubber band. For a longitudinal wave at places of compression the pressure and density tends to be maximum, while at places where rarefaction takes place, the pressure and density are minimum. Longitudinal waves are known as **Compression waves**.

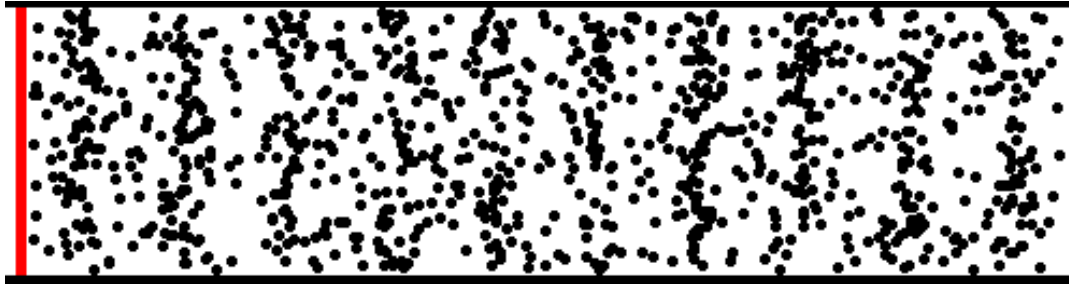
Examples of longitudinal waves : Sound waves, tsunami waves, earthquake ,P - waves, ultra sounds, vibrations in gas, and oscillations in spring, internal water waves, and waves in
slink etc.

Longitudinal Wave

wave particles vibrate back and forth along the path that the wave travels.

Compressional Wave





- Compressions
The close together part of the wave
- Rarefactions
The spread-out parts of a wave

Sound Waves:

For sound waves the existence of material medium is very necessary for the propagation of the waves. The propagation of waves taking place in solid, liquid and gases which makes us hear the sound is called **Sound wave**.

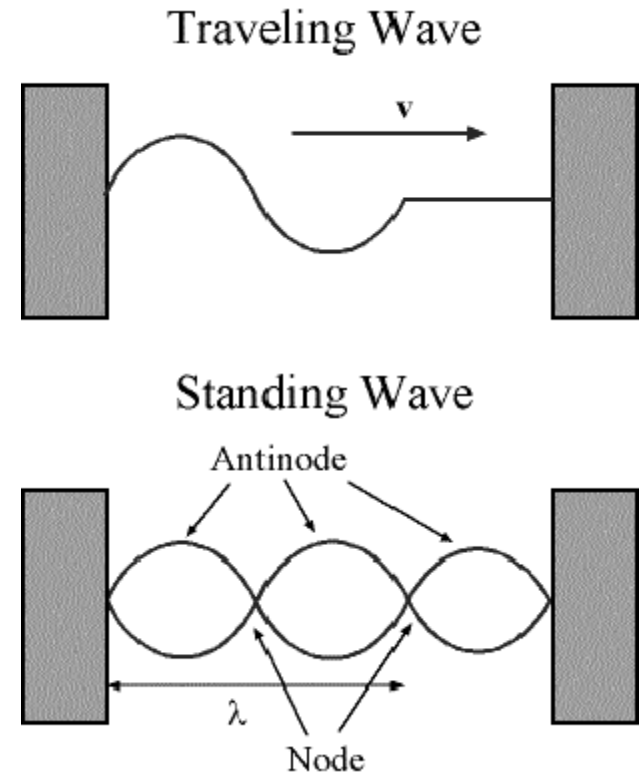
Properties :

- Sound waves are longitudinal in nature.
- Material medium is necessary for the propagation of the sound waves.
- The Speed of sound in air at N.T. P is **332 m/s**.
- The Sound is audible only between **20 Hz to 20 KHz**.

Standing Waves

A standing wave is produced when a wave that is traveling is reflected back upon itself. There are two main parts to a standing wave:

- **Antinodes** – Areas of MAXIMUM AMPLITUDE
- **Nodes** – Areas of ZERO AMPLITUDE.



Standing Waves:

When a wave remains in a constant position it is called **Standing wave**. This is possible due to 2 reasons:

- When the medium moves in a direction opposite to the direction of propagation of wave it is possible.
- When the phenomenon of interference takes place between the two waves traveling in an opposite direction then it is possible.

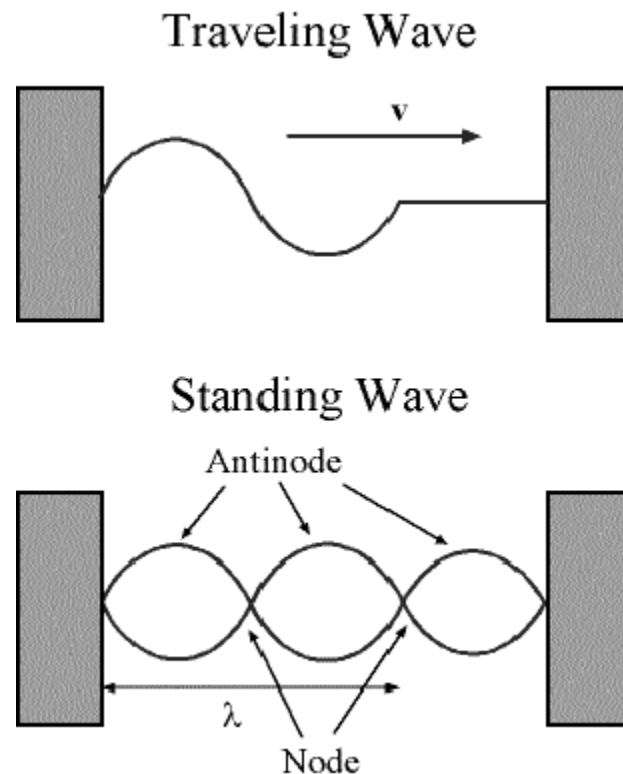
When two waves having equal frequency and amplitude overlap each other then we get a standing wave. This is possible due to the obstruction of the wave by some boundary and hence the reflection of it back in the same medium.

Standing Waves:

A standing wave is produced when a wave that is traveling is reflected back upon itself. There are two main parts to a standing wave:

Antinodes – Areas of MAXIMUM AMPLITUDE

Nodes – Areas of ZERO AMPLITUDE.



Stationary/standing waves.

Let us consider two progressive waves A and B with same amplitude, same velocity and same wavelength.

The wave A is moving with velocity V along the direction of X-axis i.e., from left to right. The wave B is moving with same velocity along the opposite direction i.e., from right to left. Then, the equation of these waves will be,

$$y_1 = a \sin \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (1)$$

$$y_2 = a \sin \frac{2\pi}{\lambda} (vt + x) \dots \dots \dots (2)$$

where, a , λ and v are the amplitude, wavelength and velocity of the waves respectively.

Due to the superposition of the two waves, the resultant displacement of the particle can be written as

$$\begin{aligned} y &= y_1 + y_2 \\ y &= a \sin \frac{2\pi}{\lambda} (vt - x) + a \sin \frac{2\pi}{\lambda} (vt + x) \\ y &= a \left[\sin \frac{2\pi}{\lambda} (vt + x) + \sin \frac{2\pi}{\lambda} (vt - x) \right] \\ &= 2a \sin \frac{2\pi}{\lambda} \cdot \frac{2vt}{2} \cdot \cos \frac{2\pi}{\lambda} \cdot \frac{2x}{2} \\ &= \left[2a \cos \frac{2\pi x}{\lambda} \cdot \sin \frac{2\pi vt}{2} \right] \\ \therefore y &= A \sin \frac{2\pi vt}{2} \dots \dots \dots (3), \end{aligned}$$

where, $A = 2a \cos \frac{2\pi x}{\lambda}$

Equation (3) is the required expression for standing waves.

Nodes: From the expression of standing waves, the amplitude is,

$$A = 2a \cos \frac{2\pi x}{\lambda}$$

This amplitude depends on the position, x of the particle. Therefore, for different positions of different particle, the values of A will be different. At points, where $A=0$ i.e., amplitude will be zero, then nodes will be created at that point.

Now, the condition for $2a \cos \frac{2\pi x}{\lambda} = 0$, will be

$$\begin{aligned}\cos \frac{2\pi x}{\lambda} &= 0 \\ \Rightarrow \frac{2\pi}{\lambda} x &= \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots \dots etc. \\ \Rightarrow x &= \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4} \dots \dots etc.\end{aligned}$$

These points are the nodes. The distance between two successive nodes will be $= \left(\frac{3\lambda}{4} - \frac{\lambda}{4} \right) = \frac{\lambda}{2}$

Therefore, the nodes are equidistant and separated by $\frac{\lambda}{2}$.

Antinodes: At which points, the resultant amplitude, A is maximum i.e., $A=\pm 2a$, antinodes will be produced at that point.

Now, the condition for $2a \cos \frac{2\pi x}{\lambda} = \pm 2a$, will be

$$\begin{aligned}\cos \frac{2\pi x}{\lambda} &= \pm 1 \\ \Rightarrow \frac{2\pi}{\lambda} x &= 0, \pi, 2\pi \dots \dots etc. \\ \Rightarrow x &= 0, \frac{\lambda}{2}, \frac{2\lambda}{2}, \frac{3\lambda}{2} \dots \dots \frac{n\lambda}{2} \cdot (n = 0, 1, 2, 3, \dots)\end{aligned}$$

These points are the antinodes. The distance between two successive antinodes will be $= \left(\frac{2\lambda}{2} - \frac{\lambda}{2} \right) = \frac{\lambda}{2}$.

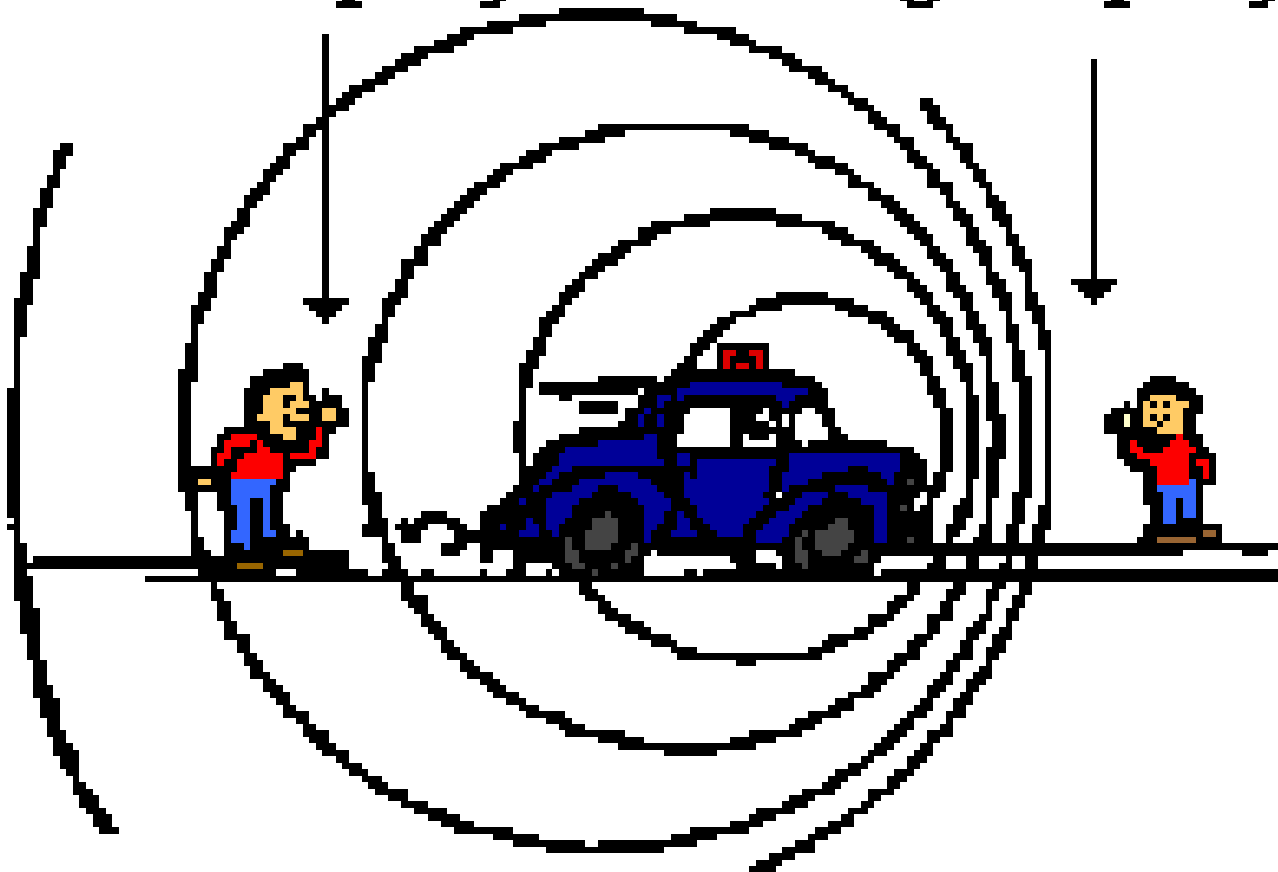
Therefore, the antinodes are equidistant and separated by $\frac{\lambda}{2}$.

Doppler Effect:

Doppler effect is the apparent change in the frequency of a sound caused by the motion of either the listener or the source of the sound.

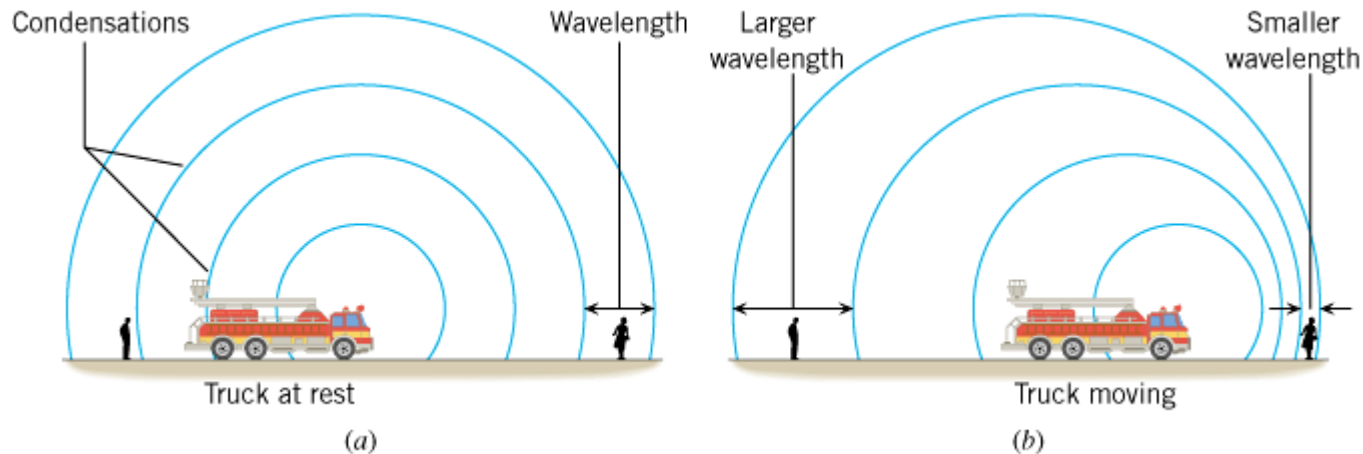
**Long Wavelength
Low Frequency**

**Small Wavelength
High Frequency**



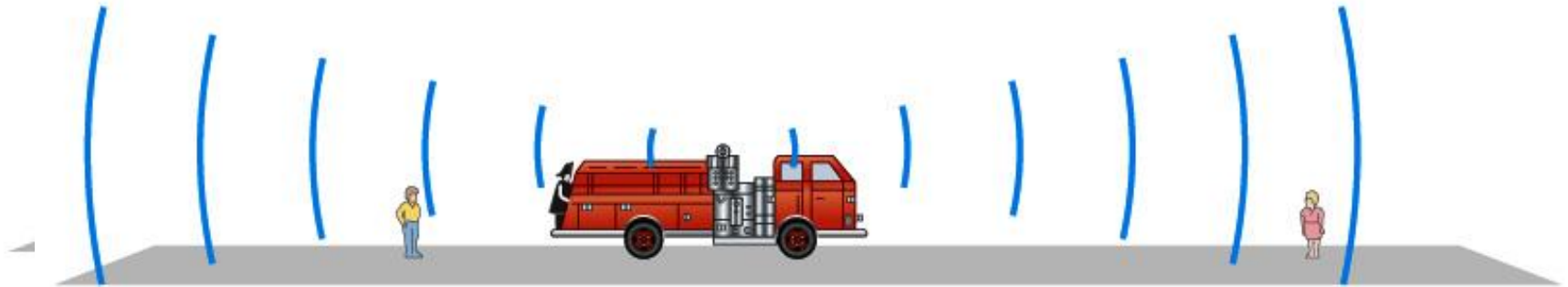
The Doppler Effect for a moving sound source

The Doppler Effect



Effect on frequency or pitch due to relative motion between the source of sound and observer/listener.

Doppler Effect



(a) At rest

8.2. Observer at Rest and Source in Motion

(a) When the source moves towards the stationary observer

Suppose a source S is producing sound of pitch n and wavelength λ . The velocity of sound is v . (Fig. 8.1).

Let the source move with a velocity a towards the observer. In one second, n waves will be contained in a length $(v - a)$ and the apparent wavelength,

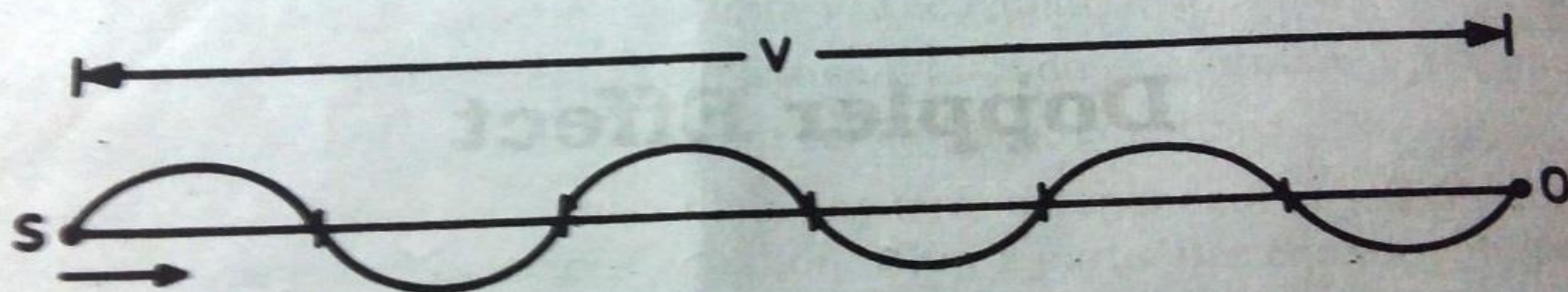
$$\lambda' = \frac{(v - a)}{n}$$

The apparent pitch,

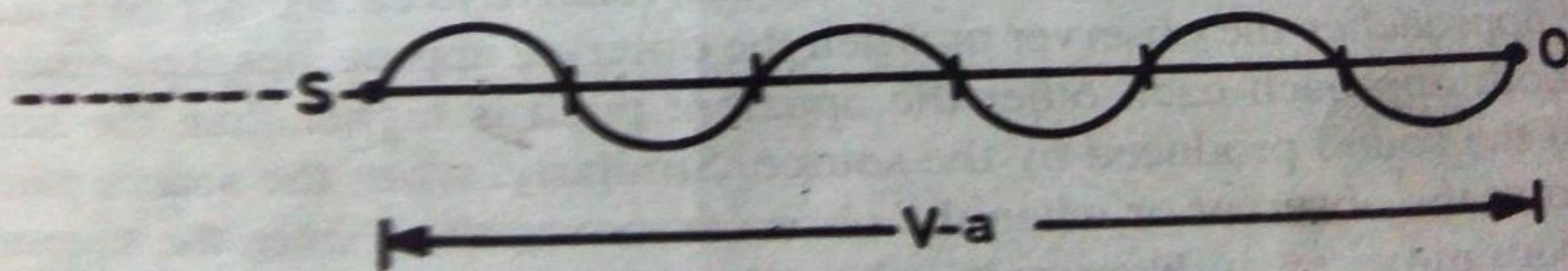
$$n' = \frac{v}{\lambda'}$$

$$\therefore n' = \left(\frac{v}{v-a} \right) n \quad \dots (1)$$

Thus, the apparent pitch of the note increases when the source moves towards a stationary observer.



(i)



(b) When the source moves away from the stationary observer

Suppose a source S is producing sound of pitch n and wavelength λ . The velocity of sound is v (Fig. 8.2). Let the source move with a velocity a away from the observer. In one second, n waves will be contained in a length $(v + a)$ and the apparent wavelength,

$$\lambda' = \frac{v + a}{n}$$

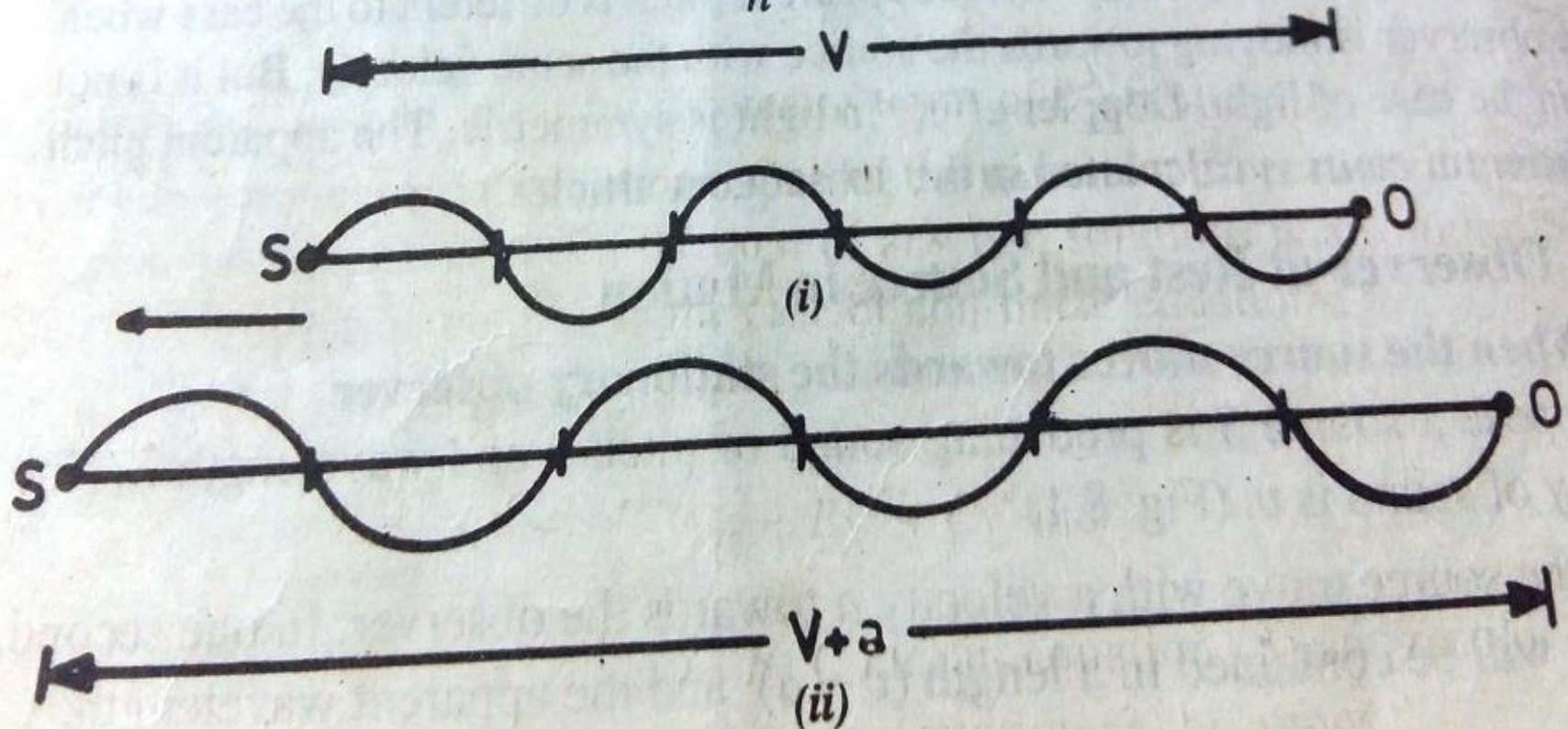


Fig. 8.2.

The apparent pitch,

$$n' = \frac{v}{\lambda'}$$

$$\therefore n' = \left(\frac{v}{v + a} \right) n \quad \dots (2)$$

Thus the apparent pitch of the note decreases when the source moves away from a stationary observer.

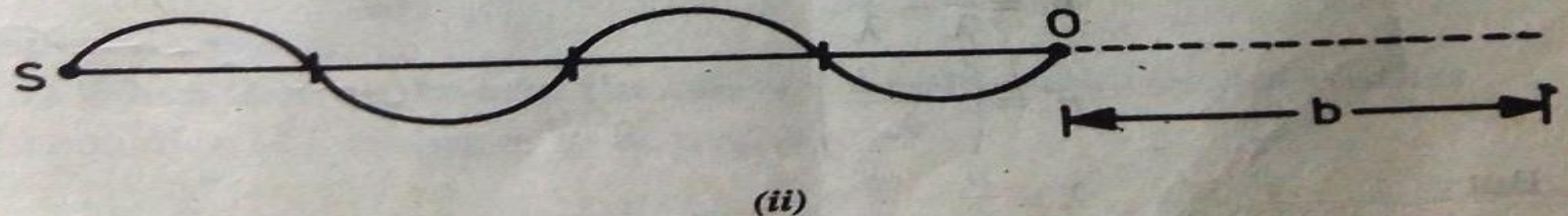
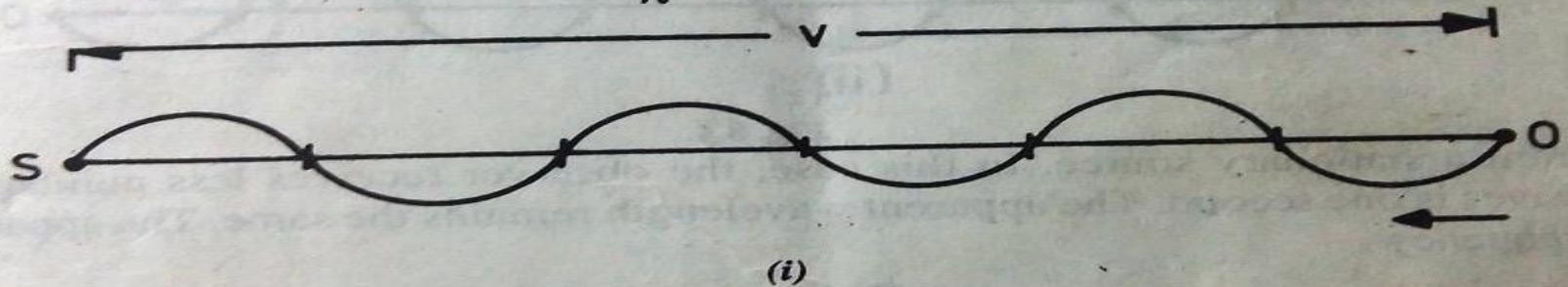
Note. For the same values of v , a and n , the values of n' in equations (1) and (2) will be different.

8.3. Source at Rest and Observer in Motion

(a) When the observer moves towards a stationary source

Suppose a source S is producing sound of pitch n and wavelength λ . The velocity of sound is v (Fig. 8.3). Let the observer move with a velocity b towards a stationary source. In this case the observer receives more number of waves in one second. The apparent wavelength remains the same. The apparent frequency

$$n' = n + \frac{b}{\lambda}$$



(ii)
Fig. 8.3.

$$n' = \frac{v}{\lambda} + \frac{b}{\lambda}$$

$$n' = \left(\frac{v + b}{\lambda} \right)$$

But

$$\lambda = \frac{v}{n}$$

$\therefore$

$$n' = \left(\frac{v+b}{v} \right) n \quad \dots (1)$$

Thus the apparent pitch of the note increases when the observer moves towards the stationary source.

towards the stationary source.

(b) **When the observer moves away from a stationary source**

Suppose a source S is producing sound of pitch n and wavelength λ (Fig. 8.4). The velocity of sound is v . Let the observer move with a velocity b away

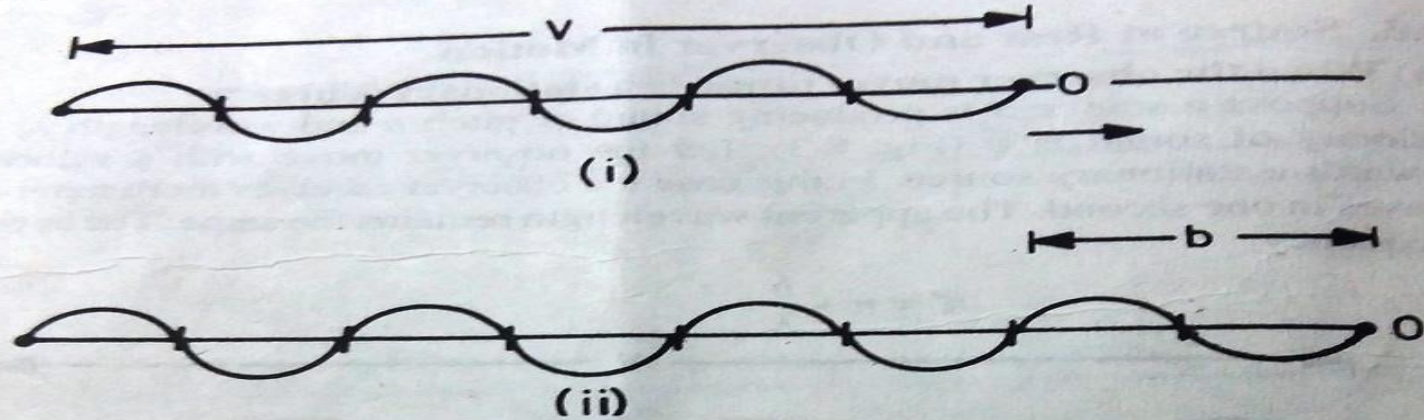


Fig. 8.4.

from a stationary source. In this case, the observer receives less number of waves in one second. The apparent wavelength remains the same. The apparent frequency

$$n' = n - \frac{b}{\lambda}$$

$$n' = \frac{v}{\lambda} - \frac{b}{\lambda}$$

$$n' = \left(\frac{v - b}{\lambda} \right)$$

But

$$\lambda = \frac{v}{n}$$

$\therefore$

$$n' = \left(\frac{v - b}{v} \right) n \quad \dots (2)$$

Thus the apparent pitch of the note decreases when the observer moves away from a stationary source.

Note. For the values of v , b and n , the values of n' in equations (1) and (2) will be different.

8.4. When both the Source and the Observer are in Motion

When the source moves towards the observer and the observer moves away from the source. Suppose a source S is producing a sound of pitch n and wavelength λ . The velocity of sound is v (Fig. 8.5). The velocity of the source is a and the velocity of the observer is b .

Let the source move towards the observer with a velocity a and the observer move away from the source with a velocity b .

The apparent wavelength

$$\lambda' = \left(\frac{v - a}{n} \right).$$

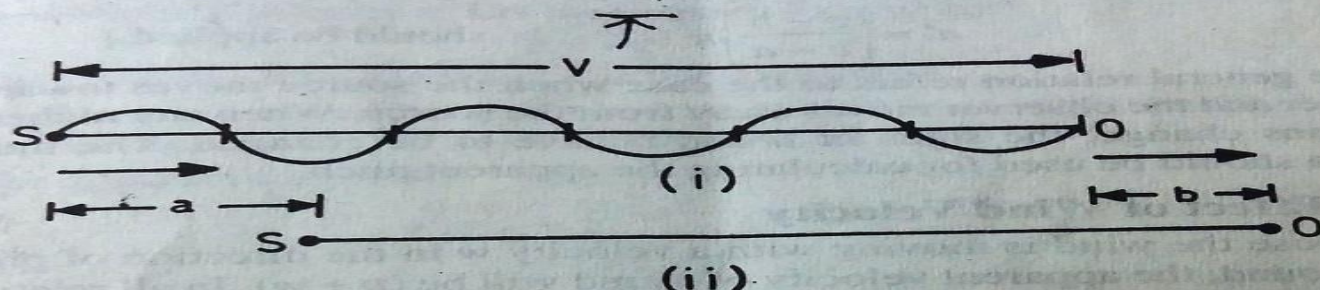


Fig. 8.5.

and

$$n' = \left(\frac{v - b}{\lambda'} \right)$$

$\therefore$

$$n' = \left(\frac{v - b}{v - a} \right) n \quad \dots (1)$$

Special cases

(a) When the source and the observer move towards each other

In equation (1), taking b to be negative,

$$n' = \left[\frac{v - (-b)}{v - a} \right] n$$

$$n' = \left(\frac{v + b}{v - a} \right) n \quad \dots (2)$$

(b) When the source and observer move away from each other

In equation (1), taking a to be negative

$$n' = \left[\frac{v - b}{v - (-a)} \right] n$$

$$n' = \left(\frac{v - b}{v + a} \right) n \quad \dots (3)$$

(c) **Source moving away from the observer and the observer moving towards the source**

In equation (1), taking both a and b negative,

$$n' = \left[\frac{v - (-b)}{v - (-a)} \right] n$$

$$n' = \left(\frac{v + b}{v + a} \right) n \quad \dots (4)$$

[Note. While solving numerical problems, the general formula

$$n' = \left(\frac{v - b}{v - a} \right) n \quad \text{should be applied.}]$$

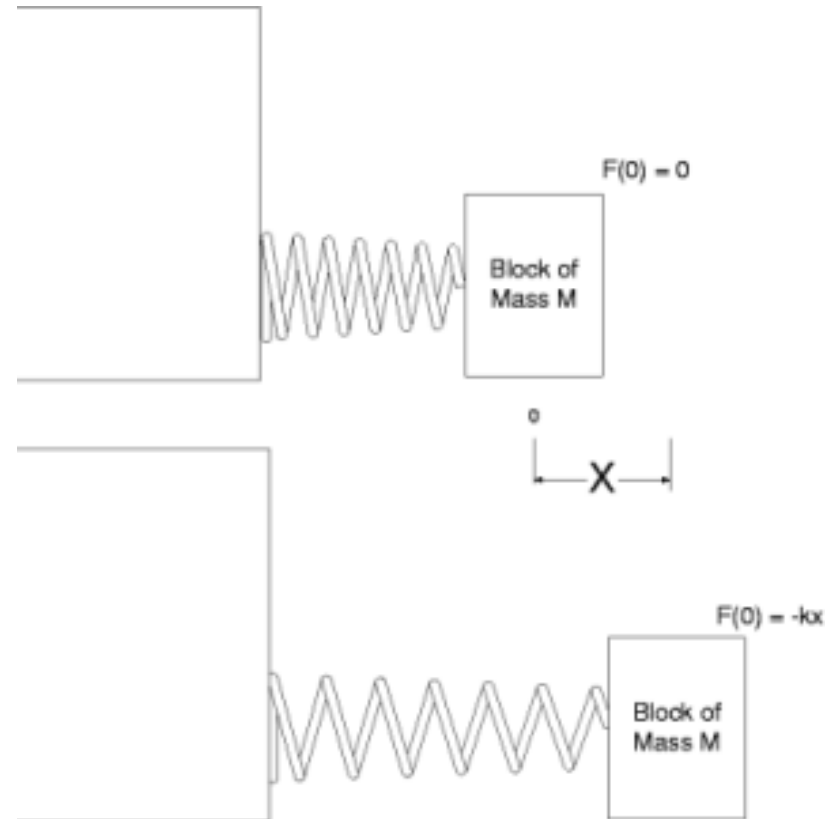
The general relation refers to the case when the source moves towards the observer and the observer moves away from the source. When any of these two directions change, the signs of a and b have to be changed. The modified formula should be used for calculating the apparent pitch.

Oscillation

Simple Harmonic Motion

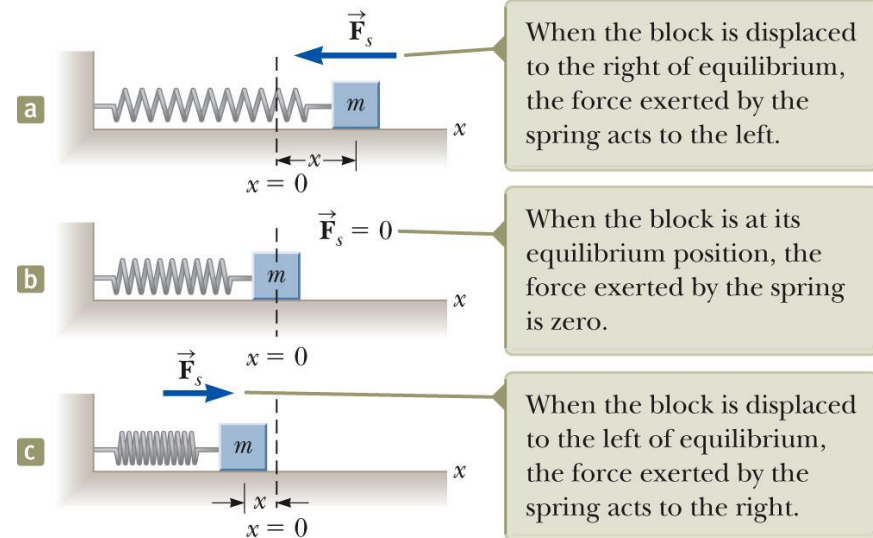
A restoring force is one that moves a system back to an equilibrium position.

Example: mass on frictionless table, attached to spring.



Periodic Motion

- Periodic motion is a motion that regularly returns to a given position after a fixed time interval.
- A particular type of periodic motion is “simple harmonic motion,” which arises when the force acting on an object is proportional to the position of the object about some equilibrium position.
- The motion of an object connected to a spring is a good example.



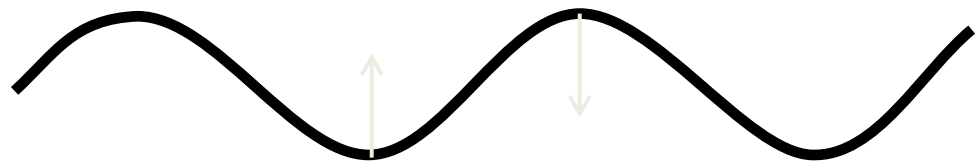
Simple Harmonic Motion:

If the acceleration of a body be proportional to its displacement from its position of equilibrium, or any other fixed point in its path and be always directed towards it, the body is said to execute a simple harmonic motion.

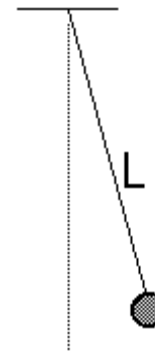
Characteristics of simple harmonic motion:

Recall Hooke's Law

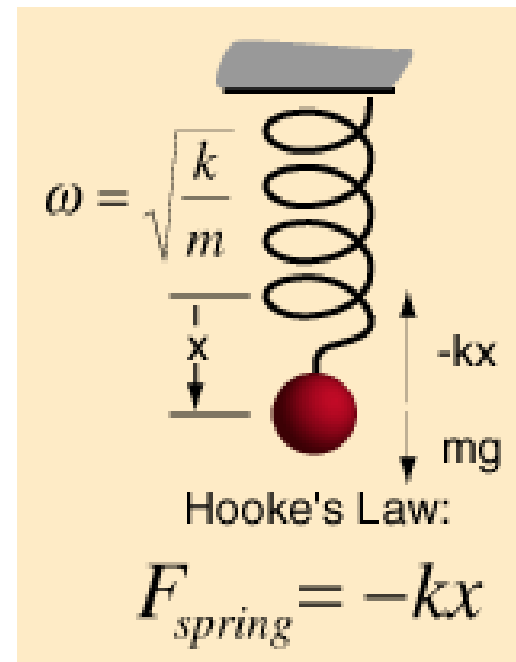
- Hooke's Law states $F_s = -kx$
 - F_s is the restoring force.
 - It is always directed toward the equilibrium position.
 - Therefore, it is always opposite the displacement from equilibrium.
 - k is the force (spring) constant.
 - x is the displacement.
- What is the restoring force for a surface water wave?



Example: gravity acting on a mass hanging from a string.



Example: gravity acting on a mass hanging from a spring.



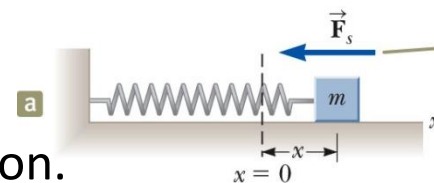
*Lots of material borrowed from:

<http://hyperphysics.phy-astr.gsu.edu/hbase/permot.html#permot>

Restoring Force and the Spring Mass System

□ In a, the block is displaced to the right of $x = 0$.

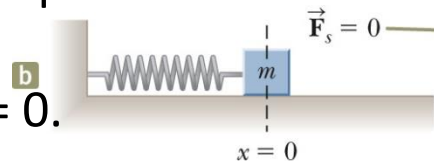
- The position is positive.
- The restoring force is directed to the left (negative).



When the block is displaced to the right of equilibrium, the force exerted by the spring acts to the left.

□ In b, the block is at the equilibrium position.

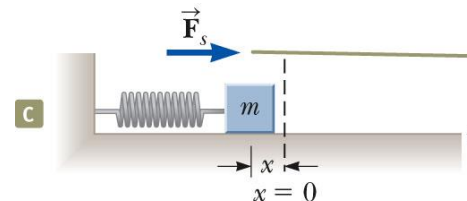
- $x = 0$
- The spring is neither stretched nor compressed.
- The force is 0.



When the block is at its equilibrium position, the force exerted by the spring is zero.

□ In c, the block is displaced to the left of $x = 0$.

- The position is negative.
- The restoring force is directed to the right (positive).



When the block is displaced to the left of equilibrium, the force exerted by the spring acts to the right.

Differential Equation of Motion

- Using $F = ma$ for the spring, we have $ma = -kx$
- But recall that acceleration is the second derivative of the position:

$$a = \frac{d^2x}{dt^2}$$

- So this simple force equation is an example of a *differential equation*,

$$m \frac{d^2x}{dt^2} = -kx \quad \text{or} \quad \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

- An object moves in simple harmonic motion whenever its acceleration is proportional to its position and has the opposite sign to the displacement from equilibrium.

Acceleration

- Note that the acceleration is NOT constant, unlike our earlier kinematic equations.
- If the block is released from some position $x = A$, then the initial acceleration is $-kA/m$, but as it passes through 0 the acceleration falls to zero.
- It only continues past its equilibrium point because it now has momentum (and kinetic energy) that carries it on past $x = 0$.
- The block continues to $x = -A$, where its acceleration then becomes $+kA/m$.

Analysis Model, Simple Harmonic Motion

- What are the units of k/m , in $a = \frac{d^2x}{dt^2} = -\frac{k}{m}x$
- They are $1/s^2$, which we can regard as a frequency-squared, so let's write it as

$$\omega^2 = \frac{k}{m}$$

- Then the equation becomes

$$a = -\omega^2 x$$

- A typical way to solve such a differential equation is to simply search for a function that satisfies the requirement, in this case, that its second derivative yields the negative of itself! The sine and cosine functions meet these requirements.

SHM Graphical Representation

- A solution to the differential equation is

$$x(t) = A\cos(\omega t + \phi)$$

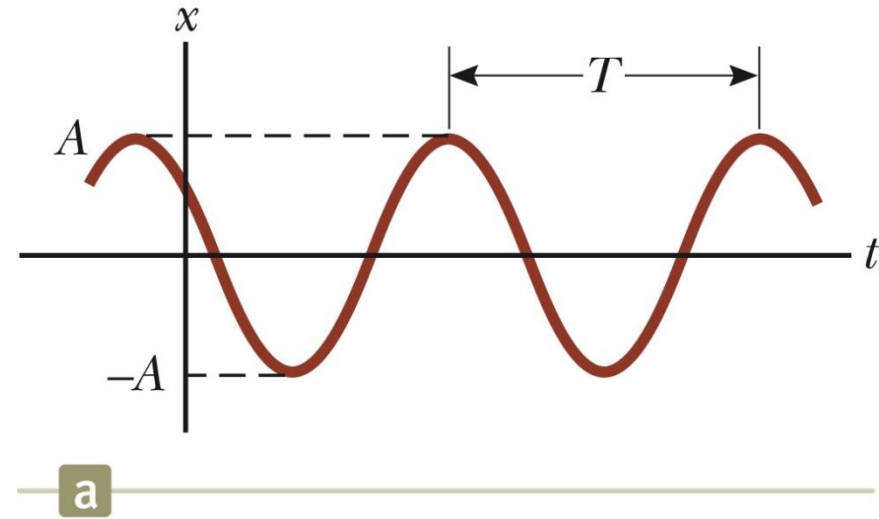
- A , ω , ϕ are all constants:

A = amplitude (maximum position in either positive or negative x direction,

ω = angular frequency, $\sqrt{\frac{k}{m}}$

ϕ = phase constant, or initial phase angle.

A and ϕ are determined by initial conditions.



Remember, the period and frequency are:

$$T = \frac{2\pi}{\omega} \quad \left(f = \frac{1}{T} = \frac{\omega}{2\pi} \right)$$

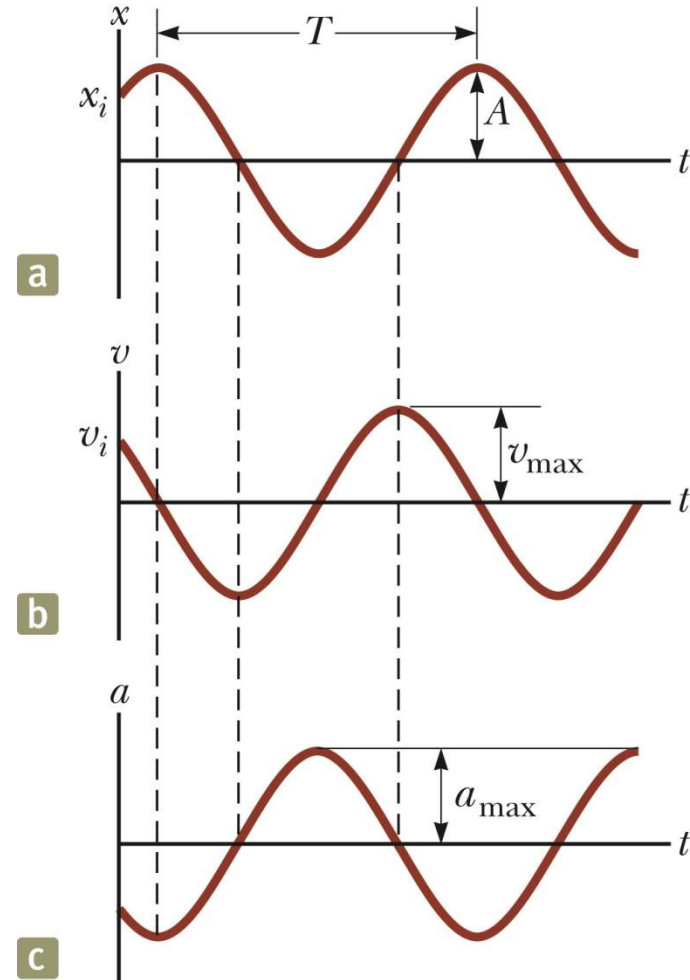
Motion Equations for SHM

$$x(t) = A\cos(\omega t + \phi)$$

$$v(t) = \frac{dx}{dt} = -\omega A\sin(\omega t + \phi)$$

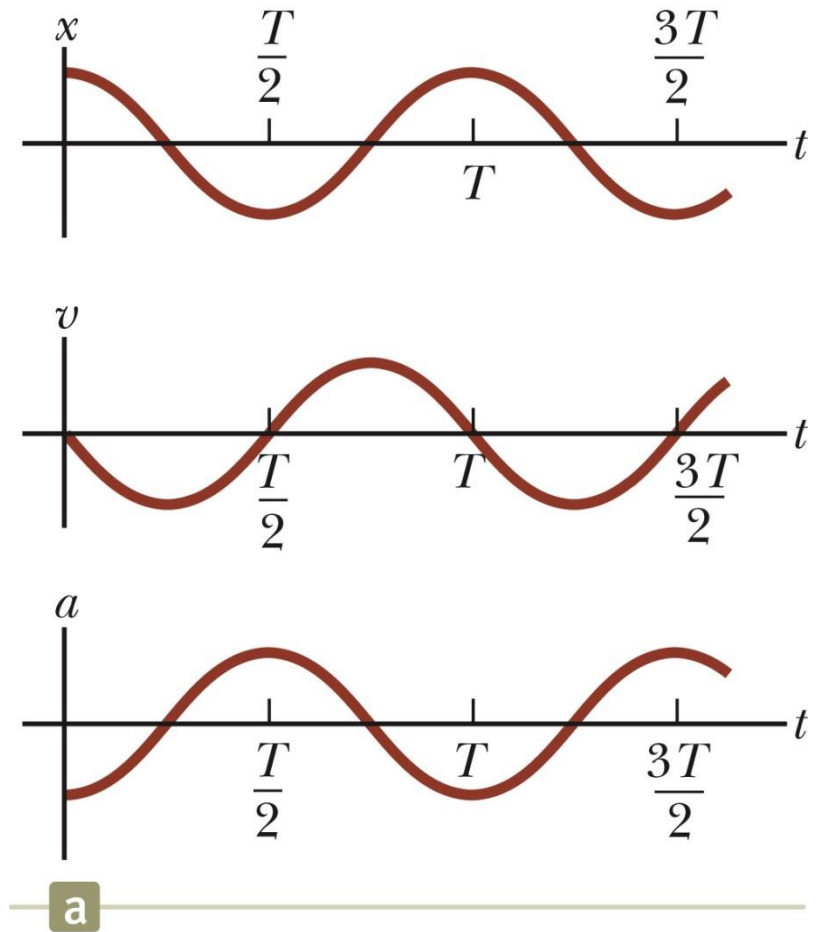
$$a(t) = \frac{d^2x}{dt^2} = -\omega^2 A\cos(\omega t + \phi)$$

The velocity is 90° out of phase with the displacement and the acceleration is 180° out of phase with the displacement.



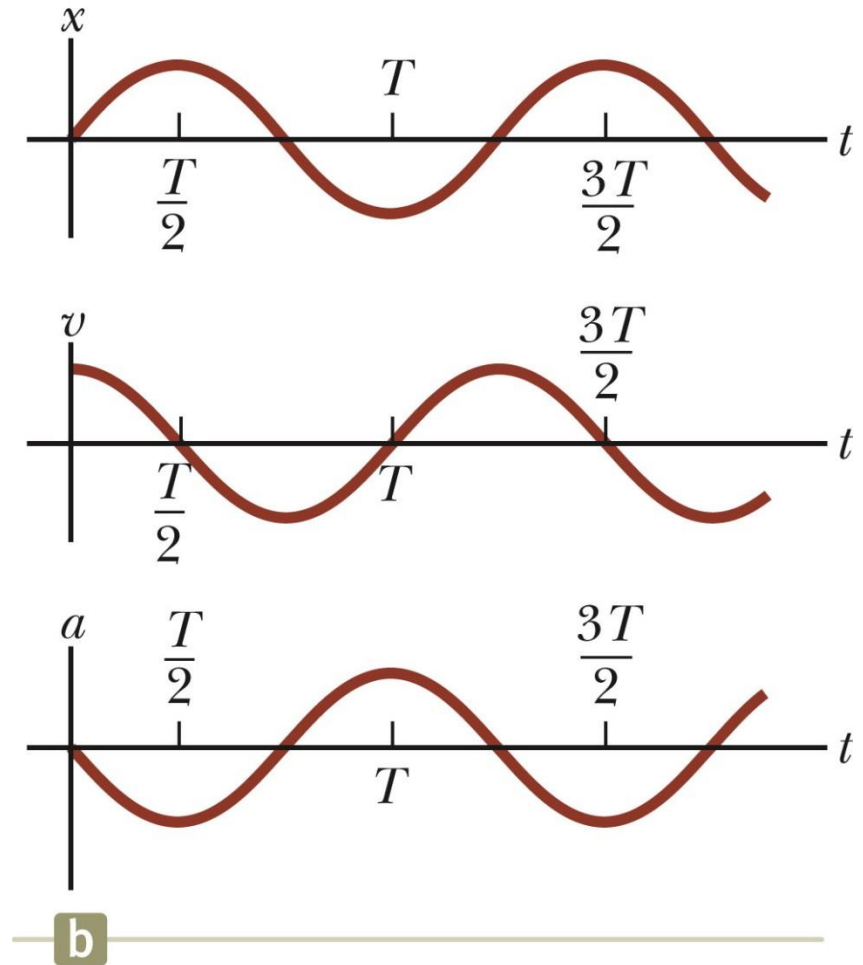
SHM Example 1

- Initial conditions at $t = 0$ are
 - $x(0) = A$
 - $v(0) = 0$
- This means $\phi = 0$
- The acceleration reaches extremes of $\pm \omega^2 A$ at $\pm A$.
- The velocity reaches extremes of $\pm \omega A$ at $x = 0$.



SHM Example 2

- Initial conditions at $t = 0$ are
 - $x(0) = 0$
 - $v(0) = v_i$
- This means $\phi = -\pi/2$
- The graph is shifted one-quarter cycle to the right compared to the graph of $x(0) = A$.



Consider the Energy of SHM Oscillator

- The spring force is a conservative force, so in a frictionless system the energy is constant
- Kinetic energy, as usual, is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \phi)$$

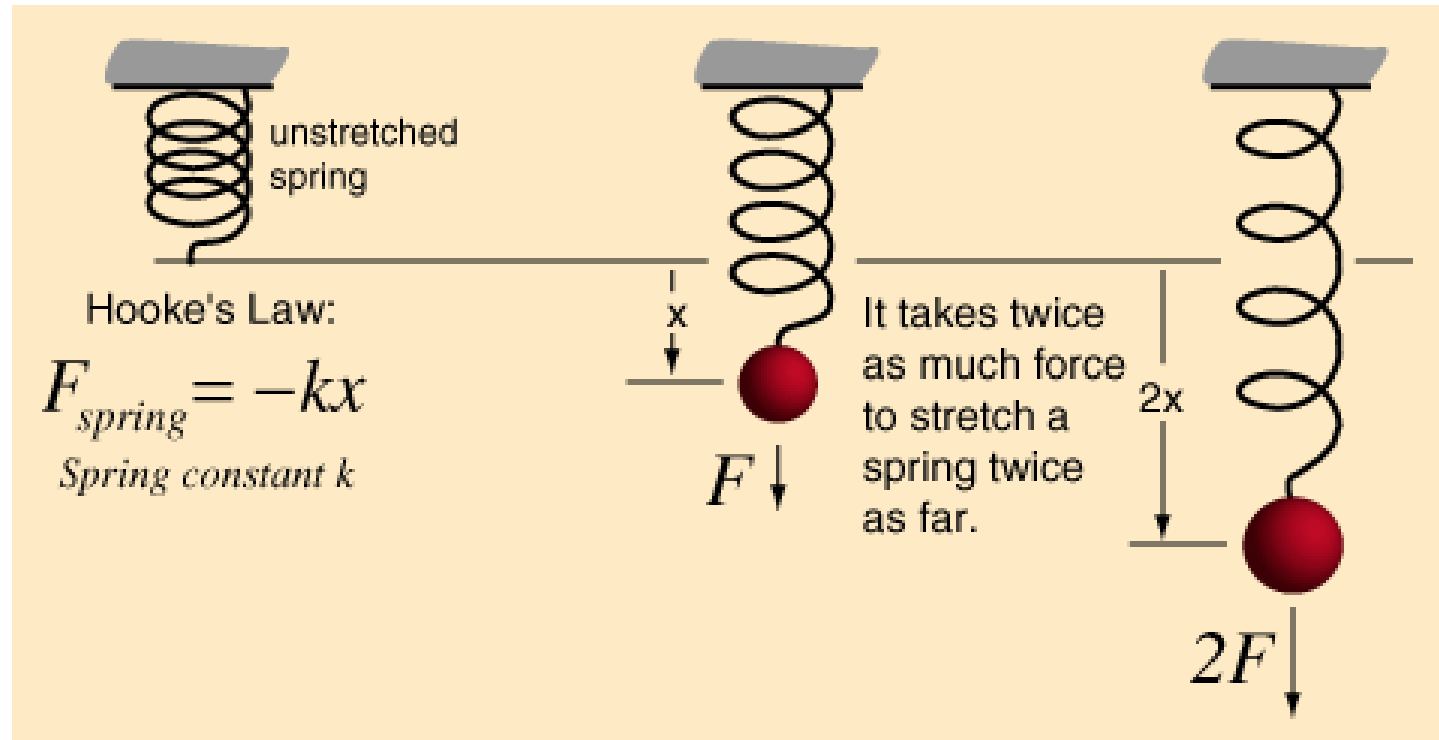
- The spring potential energy, as usual, is

$$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2(\omega t + \phi)$$

- Then the total energy is just

$$E = K + U = \frac{1}{2}kA^2 \quad (\text{a constant})$$

When the restoring force is linearly proportional to the amount of the displacement from equilibrium, the force is said to be a Hooke's Law force.



Physicists are always sloppy with their signs when doing Hooke's law calculations.

Differential Equation of SHM:

Let us consider a body of mass m attached to an ideal spring of force constant K and free to move over a frictionless horizontal surface as an example of a simple harmonic motion. If the body moving back and forth about an equilibrium position through a potential that varies as

$$U(y) = \frac{1}{2} ky^2 \dots\dots\dots (1)$$

Then the force acting on the particle is given by

$$F_{(y)} = -\frac{d}{dy}(U_y) = -\frac{d}{dy}\left(\frac{1}{2} ky^2\right)$$
$$\therefore F_{(y)} = -Ky \dots\dots\dots (2)$$

From the Newton's second law of motion, we know that,

$$F = ma$$
$$\Rightarrow -Ky = m \frac{d^2y}{dt^2} \quad [\text{From equation (2) } F = -Ky]$$
$$\Rightarrow m \frac{d^2y}{dt^2} + Ky = 0$$
$$\therefore \frac{d^2y}{dt^2} + \frac{K}{m}y = 0 \dots\dots\dots (3)$$

This is the equation of a SHM.

Let $\omega^2 = \frac{K}{m}$, then from equation (3), we have

$$\frac{d^2y}{dt^2} + \omega^2 y = 0 \dots\dots\dots (4)$$

The general solution of equation (4) is

$$y = a \sin(\omega t + \alpha) \dots\dots\dots (5)$$

where, a is the amplitude.

I

Graphical Representation of SHM:

Let P be a particle moving on the circumference of a circle of radius a . The foot of the perpendicular vibrates on the diameter YY'.

$$y = a \sin \omega t = a \sin 2\pi \frac{t}{T}$$

The displacement graph is a sine curve represented by ABCDE (Fig.-1).

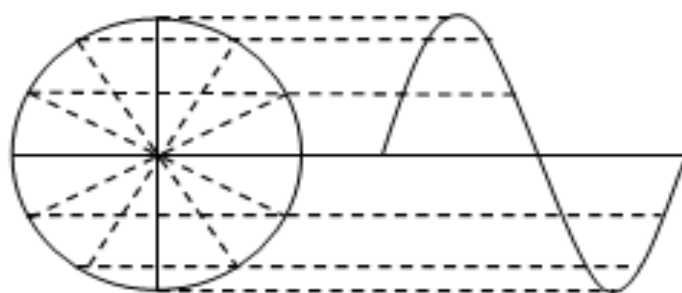


Fig.-1: displacement-time graph

The motion of the particle M is simple harmonic.

The velocity of a particle moving with simple harmonic motion is

$$v = \frac{dy}{dt} = +a\omega \cos \omega t$$

The velocity-time graph is shown in fig.-2. It is a cosine curve.

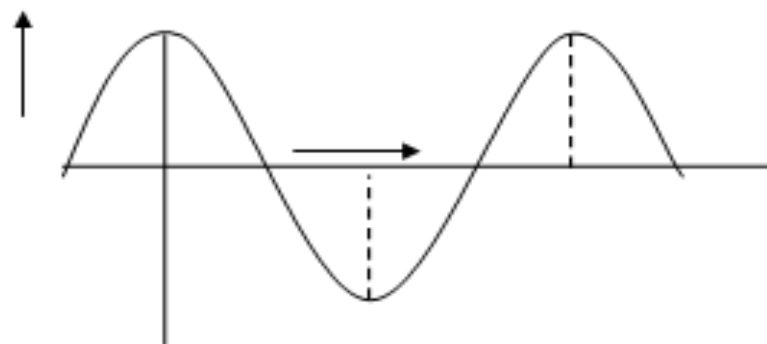


Fig.-2: velocity-time graph

The acceleration of a particle moving with simple harmonic motion is

$$\frac{d^2y}{dt^2} = -a\omega^2 \sin \omega t$$

The acceleration -time graph is shown in fig.-3. It is a negative sine curve.

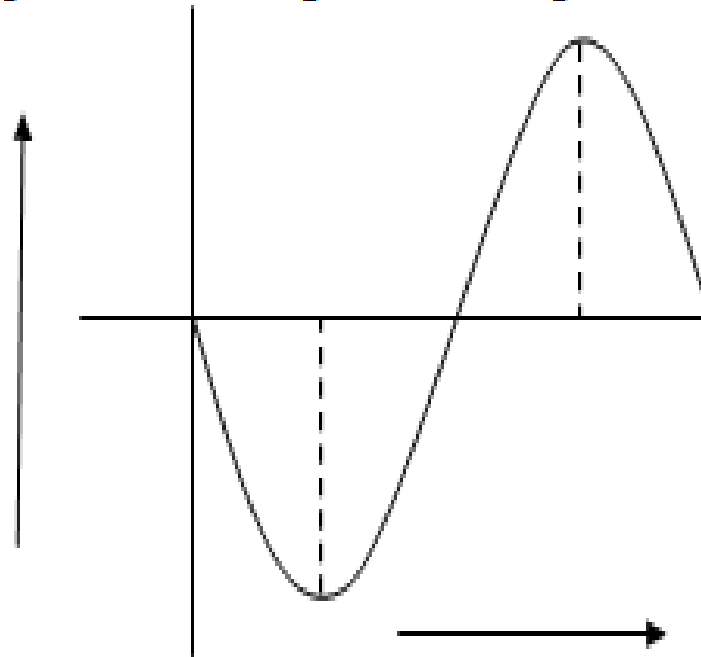


Fig.-3: Acceleration-time graph

Average Kinetic Energy of a Vibrating Particle:

The displacement of a vibrating particle is given by

$$y = a \sin(\omega t + \alpha)$$
$$v = \frac{dy}{dt} = a\omega \cos(\omega t + \alpha).$$

If m is the mass of the vibrating particle, the kinetic energy at any instant

$$= \frac{1}{2}mv^2 = \frac{1}{2}m.a^2\omega^2\cos^2(\omega t + \alpha).$$

The average kinetic energy of the Particle in one complete vibration

$$= \frac{1}{T} \int_0^T \frac{1}{2}ma^2\omega^2\cos^2(\omega t + \alpha) dt$$
$$= \frac{1}{T} \frac{ma^2\omega^2}{4} \int_0^T 2\cos^2(\omega t + \alpha) dt$$
$$= \frac{ma^2\omega^2}{4T} \int_0^T [1 + \cos 2(\omega t + \alpha)] dt$$
$$= \frac{ma^2\omega^2}{4T} \left[\int_0^T dt + \int_0^T \cos 2(\omega t + \alpha) dt \right]$$

$$\text{But, } \int_0^T \cos 2(\omega t + \alpha) dt = 0$$

$$\therefore \text{Average K.E.} = \frac{ma^2\omega^2}{4T} \cdot T + 0$$

$$= \frac{ma^2\omega^2}{4} = \frac{ma^2(4\pi^2n^2)}{4} = \pi^2ma^2n^2$$

where m is the mass of the vibrating particle, a is the amplitude of vibration and n is the frequency of vibration. Also, the average kinetic energy of a vibrating particle is directly proportional to the square of the amplitude.

Total Energy of a Vibrating Particle

$$\begin{aligned}y &= a \sin(\omega t + \alpha) \\ \sin(\omega t + \alpha) &= \frac{y}{a} \\ \cos(\omega t + \alpha) &= \sqrt{1 - \frac{y^2}{a^2}} = \sqrt{\frac{a^2 - y^2}{a^2}} \\ &= \frac{\sqrt{a^2 - y^2}}{a}\end{aligned}$$

$$\text{Velocity } v = a\omega \cos \omega t = a\omega \frac{\sqrt{a^2 - y^2}}{a} = \omega \sqrt{(a^2 - y^2)}$$

$\therefore$ The kinetic energy of the Particle at the instant the displacement is y ,

$$\begin{aligned}&= \frac{1}{2}mv^2 \\ &= \frac{1}{2}m \cdot \omega^2 (a^2 - y^2)\end{aligned}$$

Potential energy of the vibrating particle is the amount of work done in overcoming the force through a distance y .

$$\text{Acceleration} = -\omega^2 y$$

$$\text{Force} = -m\omega^2 y$$

(The $-ve$ sign shows that the direction of the acceleration and force are opposite to the direction of the motion of the vibrating particle.)

$$\begin{aligned}\therefore \text{P.E.} &= \int_0^y m \cdot \omega^2 y \cdot dy \\ &= m\omega^2 \cdot \frac{y^2}{2} = \frac{1}{2} m\omega^2 y^2.\end{aligned}$$

Total energy of the particle at the instant the displacement is y

$$\begin{aligned}&= \text{K.E.} + \text{P.E.} \\ &= \frac{1}{2}m \cdot \omega^2 (a^2 - y^2) + \frac{1}{2} m\omega^2 y^2 \\ &= \frac{1}{2} m\omega^2 \cdot a^2 \\ &= \frac{1}{2} m(2\pi n)^2 \cdot a^2 = 2\pi^2 ma^2 n^2.\end{aligned}$$

As the average kinetic energy of the vibrating particle $= 2\pi^2 ma^2 n^2$, the average potential energy $= \pi^2 ma^2 n^2$. The total energy at any instant is a constant.

Free Vibrations:

When the bob of a simple pendulum (in vacuum) is displaced from its mean position and left, it executes simple harmonic motion. The time period of oscillation depends only on the length of the pendulum and the acceleration due to gravity at the place. The pendulum will continue to oscillate with the same time period and amplitude for any length of time. In such case there is no loss of energy by friction or otherwise. In all similar cases, the vibrations will be undamped free vibrations. The amplitude of swing remains constant.

Damped Oscillation:

In actual practice, when the pendulum vibrates in air medium, there are frictional forces and consequently energy is dissipated in each vibration. The amplitude of swing decreases continuously with time and finally the oscillations die out. Such vibrations are called free damped vibrations.

In a word, in the presence of frictional or viscous forces, the amplitude of vibration decreases continuously with time and finally the oscillations die out. Such vibrations are called free damped vibrations.

Forced Oscillation:

The time period of a body executing simple harmonic motion depends on the dimensions of the body and its elastic properties. The vibrations of such a body die out with time due to dissipation of energy. If some external periodic force is constantly applied on the body, the body continues to oscillate under the influence of such external forces. Such vibrations of the body are called forced vibrations.